The BIND Lightcone Suite IV: Gas Thermodynamics Across a 30-Parameter Feedback Space, Confronted with DESI×ACT kSZ and eROSITA

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Abstract

Recent stacked kinematic Sunyaev-Zel'dovich (kSZ) measurements from DESI×ACT and eROSITA X-ray group statistics have been read as evidence that circumgalactic and intragroup gas is more efficiently evacuated by feedback than any single IllustrisTNG realization predicts. Because that comparison uses one simulation against one (or a few) subgrid feedback parametrizations, it cannot distinguish a genuine tension with the TNG galaxy-formation *model* from an unfortunate choice of TNG feedback *parameters*. We remove that ambiguity by painting the BIND conditional flow-matching emulator's hydrodynamical fields onto the *same* $10^{13}\,\mathrm{M}_{\odot}/h$ halos of a TNG300 dark-matter-only lightcone under a 256-node Sobol sampling of the 30-dimensional IllustrisTNG feedback prior, producing an 8.6M-halo-instance integrated catalog together with compensated-aperture-photometry (CAP) stacks of the electron column $\tau = \sigma_T \int n_e\, dl$ and the Compton-y parameter. We show that *no* point in this 256-node design reaches the eROSITA strong-feedback group gas-fraction band at $M_{500c} \sim 10^{13.5}\,\mathrm{M}_{\odot}/h$ (0% coverage, including the 2.5% strongest-feedback tail), reproducing the Siegel/Bigwood missing-baryon tension across a continuous parameter volume rather than a single point. Against the exact DESI×ACT σ_v-free CAP-ratio observable, the fiducial run is 1.76×/1.40× too gas-rich (stellar-mass cuts $\log M_\star$>11.0/11.25) but the tension is feedback- and data-limited, not absolute: roughly 20% of the 256 nodes are statistically consistent with the data within its errors. A CMB-lensing κ mass anchor shows BIND's halo masses are accurate to ∼30%, confirming the discrepancy is a gas-content, not a mass-selection, effect. The 30-dimensional (τ, y) feedback response collapses onto a clean, physically interpretable 2-D latent manifold – an independent inner-gas ($f_{\rm gas}(< M_{500c})$) axis and an outer-gas ($f_{\rm gas}(M_{500c} \to M_{200c})$) axis, each correlating with its latent direction at $r = 0.95$ – and a companion ray-traced, field-level analysis shows this splits into just one dominant gas-ejection direction once projected through the lensing kernel. We state plainly and repeatedly that BIND's "kSZ" is this velocity-free optical depth, not a reconstructed temperature decrement, and discuss what a genuine $\Delta T_{\rm kSZ}$ comparison would require. The paper's central claim is that the TNG-like feedback *model space*, not any single parameter choice within it, is what is in tension with the data.

1 Introduction

Stacked kinematic Sunyaev-Zel'dovich (kSZ) measurements have become one of the sharpest available probes of the circumgalactic and intragroup gas that hydrodynamical galaxy-formation simulations struggle to place correctly. Pairwise and stacked kSZ analyses around spectroscopic galaxy samples (Schaan et al., 2021; Amodeo et al., 2021) measure the velocity-weighted free-electron column along the line of sight, and because the kSZ signal is (to leading order) insensitive to the gas temperature, it provides a comparatively clean tracer of where baryons *are*, independent of how hot they are. The combination of DESI spectroscopy with ACT DR6 CMB maps has pushed this technique to a large, well-characterized sample of luminous red galaxies (LRGs), bright galaxy sample (BGS) hosts, and emission-line galaxies (ELGs) (Qu et al., 2026; Hadzhiyska et al., 2026). In parallel, eROSITA X-ray group and cluster statistics have reported hot-gas fractions at group scales that sit systematically below both kSZ-inferred values and standard hydrodynamical-simulation predictions (Popesso et al., 2024). Read together, these results have been interpreted as evidence for a "missing baryons" or "over-cooled/under-ejected" tension: real group- and cluster-scale gas appears to be more efficiently redistributed by feedback than IllustrisTNG-like simulations predict (Siegel et al., 2025; Kovač et al., 2025; Bigwood et al., 2024).

The trouble with this comparison, as usually made, is that it pits one survey against one simulation realization – typically the fiducial IllustrisTNG run, with one fixed choice of the ∼30 subgrid feedback parameters that govern stellar and AGN winds, black-hole accretion and radio-mode feedback, and the initial mass function. A single-point comparison of this kind cannot distinguish two qualitatively different failure modes: (i) the TNG galaxy-formation *model* – its physical ingredients and functional forms – is simply incapable of reproducing the observed gas distribution at any parameter setting, or (ii) the TNG *model* is adequate but the particular parameter values used in the public IllustrisTNG runs happen to sit in the wrong part of a otherwise-viable parameter space. These two possibilities call for entirely different remedies – new physics versus recalibration – and only a comparison against the *full* feedback prior volume, not a single realization, can tell them apart.

This paper closes that gap using the BIND emulator (Paper I, in prep., Lee, 2026), a conditional flow-matching model that paints IllustrisTNG-like hydrodynamical fields onto dark-matter-only (DMO) halos given a joint cosmological-plus-astrophysical parameter vector. We paint the *same* population of TNG300-Dark lightcone halos 256 times, once per node of a Sobol sampling of the 30-dimensional IllustrisTNG feedback prior (holding cosmology fixed at the fiducial value), so that every difference between nodes is attributable to feedback parameters alone and not to sample variance, resolution, or halo-finder differences. We then confront this continuous feedback ensemble with the DESI×ACT kSZ CAP profiles and the eROSITA group gas-fraction relation using the exact observables the surveys report, and ask a single question: does *any* point in the 256-node design match the data? Section 2 describes the suite, the per-halo integrated catalog, the CAP-filter machinery, and states plainly that BIND's "kSZ" observable is the velocity-free electron-column optical depth, not a literal temperature decrement. Section 3 presents the eROSITA confrontation, the CAP-ratio gas-fraction confrontation, the resulting first continuous-feedback kSZ+tSZ posterior, and the 2-D feedback latent that the per-halo response collapses onto, together with a ray-traced field-level companion analysis. Section 4 argues that the pattern of results supports a model-space rather than a parameter-space interpretation of the tension, and Section 5 summarizes.

2 Methods

2.1 Suite recap: the same halos, 256 feedback nodes

We use the BIND suite described in Paper I (Lee, 2026, in prep.): a 256-node Sobol design over the 30 IllustrisTNG astrophysical feedback parameters (stellar and AGN wind/radio feedback, black-hole accretion and seeding, the IMF slope, supernova rates, and related knobs), each node painted by the BIND conditional flow-matching emulator onto the *same* set of dark-matter halos drawn from a TNG300-Dark ($L = 205\,\mathrm{Mpc}/h$) lightcone, ray-traced with the project's `lux` pipeline to produce tomographic convergence (κ), electron-column (τ), and Compton-y maps. Of the 256 Sobol nodes, 253 are valid/usable for the ray-traced field-level products used in Section 3.6[TODO: reason for the 3/256 field-level exclusions is not recorded in the plan doc or notebook; trace before submission]. The per-halo (non-ray-traced) products used throughout Sections 3.1–3.5 draw on all 256 nodes, with a small number excluded per-analysis by finite/NaN masks (e.g. where a CAP stack or a GNFW fit fails to converge for a given node). Because every node shares the identical DMO lightcone, halo positions, masses, and merger histories, differences between nodes are attributable to the painted feedback response alone.

2.2 The per-halo integrated catalog

For the per-halo (non-ray-traced) analyses we use an integrated catalog built by aggregating spherical-aperture quantities around every halo in every Sobol run and snapshot: 8.6M halo-instance rows (halo $\times$ Sobol node $\times$ snapshot) $\times$ 55 columns, including f_{gas} and total/gas/stellar/dark-matter mass at both $200c$ and $500c$ overdensities, the integrated Compton-Y parameter and mass-weighted temperature at $200c/500c$, M_{200c}, r_{200c}, redshift, scale factor, and all 30 astrophysical parameters of the generating node. This catalog is the basis for the eROSITA f_{gas}–mass confrontation (Section 3.1) and the σ_v-free 3D gas-fraction estimates (Section 3.3).

2.3 CAP filter stacks

The exact DESI$\times$ACT kSZ and tSZ observable is compensated aperture photometry (CAP): a filter that differences a disk of aperture radius θ_d ($+1$ weight, $\theta < \theta_d$) against an equal-area surrounding ring (-1 weight, $\theta_d < \theta < \sqrt{2}\,\theta_d$), which removes the mean background and any constant offset along the line of sight (Hadzhiyska et al., 2026). We reproduce this filter exactly (Figure 2, panel c) and apply it to the painted τ and y maps stacked in physical apertures matched to the DESI host samples. Following the survey's own convention, the reported observable is the CAP-*ratio* $\tilde{f}_{\mathrm{gas}}(\theta_d) \equiv [\mathrm{CAP}_{\mathrm{gas}}(\theta_d)/\mathrm{CAP}_{\mathrm{mat}}(\theta_d)]\,/(\Omega_b/\Omega_m)$ (Hadzhiyska et al., 2026), i.e. the CAP value of the gas divided by the CAP value of the total matter at the same aperture, itself normalized by the cosmic baryon fraction so that $\tilde{f}_{\mathrm{gas}} = 1$ corresponds to the cosmic value. This is a *compensated*, not cumulative, aperture ratio – a distinction that matters in practice (Section 2.8).

2.4 GNFW profile comparison

We additionally compare the shape of the stacked $\tau(R)$ profile to the generalized NFW (GNFW) fits reported for the DESI$\times$ACT sample (Hadzhiyska et al., 2026), of the form

$$\rho_{\mathrm{gas}}(r) = f_b\,\rho_{\mathrm{cr}}(z)\,\rho_0 \left(\frac{r}{x_c r_{200c}}\right)^{\gamma} \left[1 + \left(\frac{r}{x_c r_{200c}}\right)^{\alpha}\right]^{-(\beta+\gamma)/\alpha}, \tag{1}$$

with fixed inner slope $\gamma = -0.5$ and core scale $x_c = 0.7$, and free $\{\rho_0, \alpha, \beta\}$, forward-projected along the line of sight (no Abel inversion) to a model $\tau(R/r_{200c})$ for direct comparison to the painted stacks; results of this comparison are presented in Section 3.2.

2.5 A σ_v-free gas-fraction estimator

Converting a measured CAP temperature decrement into a physical optical depth or gas mass formally requires the survey's pairwise-velocity dispersion σ_v and its reconstruction fidelity, which fold in redshift-space distortions, photometric-redshift scatter, and the survey's own velocity calibration. We bypass this step entirely wherever possible by working with a *ratio* observable, $\tilde{f}_{\rm gas} = f_{\rm gas}/(\Omega_b/\Omega_m)$, computed two ways: (a) directly, in 3D, from the per-halo integrated catalog (no projection, no σ_v), and (b) as a continuous projected curve from the painted patches, $\Sigma M_{\rm gas}(< R)/\Sigma M_{\rm tot}(< R)/(\Omega_b/\Omega_m)$, matched in aperture to the CAP-ratio definition above. Both versions are reported below and are labelled distinctly, since they are not the same observable (Section 3.3).

2.6 Posterior machinery

To turn the 256-node design into a feedback posterior we train a Gaussian-process emulator (ARD Matérn kernel, `scikit-learn`, joblib-parallel hyperparameter search) mapping the 30 feedback parameters to the CAP observable(s) at each aperture, then sample the resulting population likelihood – built from the CAP stacks over the relevant DESI host bins – against the real (or, for forecasts, mock) data and its covariance with `emcee`, under the flat Sobol-box prior.

2.7 bind's "kSZ" is the velocity-free electron column

We state this plainly because it shapes every comparison in this paper: BIND does not carry a gas velocity field, and none of the quantities labelled "kSZ" anywhere in this work are a reconstructed temperature decrement $\Delta T_{\rm kSZ} = -(\sigma_T/c)\int n_e v_r\, dl$. What we compute and paint is the velocity-free electron-column optical depth $\tau = \sigma_T \int n_e\, dl$ (equivalently, the same line integral that gives a fast-radio-burst dispersion measure). Every amplitude comparison in this paper therefore enters through the CAP-ratio $\tilde{f}_{\rm gas}(\theta)$ built from τ, never through an attempted reconstruction of $\Delta T_{\rm kSZ}$. Building a genuine $\Delta T_{\rm kSZ}$ observable would require a dark-matter-only velocity surrogate for the (currently unmodelled) gas peculiar-velocity field; we return to this in Section 4.

2.8 Methods-lesson footnote: a mass-convention bug, and two normalization pitfalls

We flag three methodological missteps encountered and corrected during this analysis, not as results but because each one produced a spurious "tension" number that circulated before being caught, and because the lessons generalize to any future BIND×survey comparison.

First, the tSZ y-CAP leg originally selected LRG hosts using a stellar-to-halo-mass relation band of $\log M_{200c} \in [13.2, 13.5]$ in $\mathrm{M}_\odot/h$, but the DESI-LRG host mass inferred from ACT CMB lensing is $\log M_{200c} = 13.18\,\mathrm{M}_\odot/h$ (Sailer et al., 2024), so the original selection was ∼0.2 dex too high in host mass. Because the integrated Compton-Y parameter scales steeply with mass ($Y \propto M^{5/3}$), this ∼0.2 dex offset alone inflated the derived y-CAP amplitude by a spurious factor of 2.6×. We corrected the host-mass band to $\log M_{200c} \in [13.0, 13.35]\,\mathrm{M}_\odot/h$ and switched from a median to a mean stack, and use only the corrected value throughout (Section 3.5). This bug is a methods lesson, not a result: any inter-probe (kSZ-vs-tSZ) "tension" statement made before this fix should be discarded, as we discuss further in Section 3.5.

Second, comparing BIND's *local* $\tau(R)$ directly to the forward-projected GNFW fit (Section 2.4) produced a spurious ∼30× "tension" before we recognized that the DESI×ACT data point is $T_{\rm kSZ}^{\rm CAP}$, the compensated-aperture signal in arcmin^2, a categorically different observable from a local $\tau(R)$ profile value; the tabulated GNFW ρ_0 was also under-normalized for a volume integral. Only CAP-filtered, aperture-matched quantities are compared to data anywhere in this paper.

Third, even after adopting the CAP filter, using a *cumulative* (rather than compensated) aperture for $\tilde{f}_{\rm gas}$ retains roughly 50 Mpc/h of uncancelled line-of-sight background that the true CAP

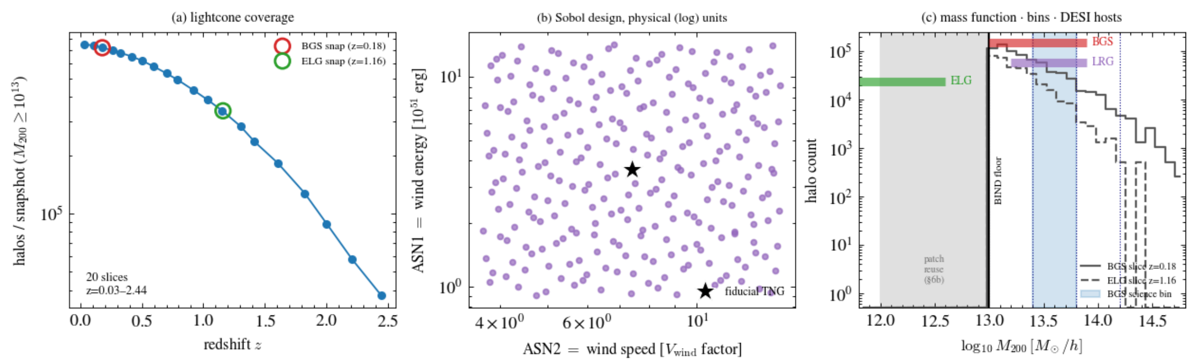

Figure 1: **(a)** Number of $M_{200c} \geq 10^{13}\,\mathrm{M}_\odot/h$ halos per lightcone snapshot slice across the 20 snapshots spanning $z = 0.034$–2.444, with the BGS ($z = 0.18$) and ELG ($z = 1.16$) confrontation slices marked. **(b)** The 256-node Sobol feedback design in the physical units of its two dominant wind parameters (wind energy, wind speed factor), with the fiducial IllustrisTNG point starred. **(c)** The halo mass function at the BGS and ELG slices relative to the BIND painted-halo mass floor of $M_{200c} = 10^{13}\,\mathrm{M}_\odot/h$ and the DESI BGS/LRG/ELG host-mass bands; the shaded region below the floor is recovered only via the patch-reuse technique (Section 4.1). Takeaway: the design spans the full 30-parameter feedback prior on an identical halo population across the full observable lightcone.

ratio is designed to remove; this manufactured an earlier, incorrect "0 of 256 Sobol nodes reach the data" claim that we explicitly retract (see the boxed note in Section 3.3). We additionally found and fixed an angular-diameter-distance bug (a spurious extra factor of $1/h$ in $D_A(z)$, inflating D_A by $1.476\times$) that had shrunk the derived aperture $\theta(r_{200})$ and biased the inferred data $\tilde{f}_{\rm gas}(r_{200})$ low, from an erroneous 0.28 to the corrected value of 0.44–0.60 used throughout this paper.

3 Results

3.1 Gas fraction versus mass: 0% coverage of the eROSITA strong-feedback band

Figure 1 summarizes the suite: the redshift coverage of the lightcone, the physical-unit Sobol design over the two dominant wind parameters, and the halo mass function relative to the DESI host bands and the BIND painted-halo floor. Figure 2 shows the fiducial $\tau(R/r_{200c})$ stack by mass bin, the stellar-to-halo-mass matching used to place DESI $M_\star$ cuts onto M_{200c}, and the CAP filter itself.

Our headline gas-content result is shown in Figure 3: at $M_{500c} \sim 10^{13.5}\,\mathrm{M}_\odot/h$, the entire 256-node Sobol feedback band sits at $f_{\rm gas} \approx 0.11$–0.15, and *zero* of the 256 nodes reach the eROSITA/Popesso et al. (2024) strong-feedback gas-fraction band, even at the 2.5%-strongest-feedback edge of the design. This reproduces the Siegel/Bigwood missing-baryon tension (Siegel et al., 2025; Bigwood et al., 2024) across a continuous 30-dimensional feedback parameter space, rather than a single simulation realization, and is the paper's central, most robust result: it was independently reproduced across the D1 reduction pass and is repeated without qualification in every later revision of the analysis notebook.

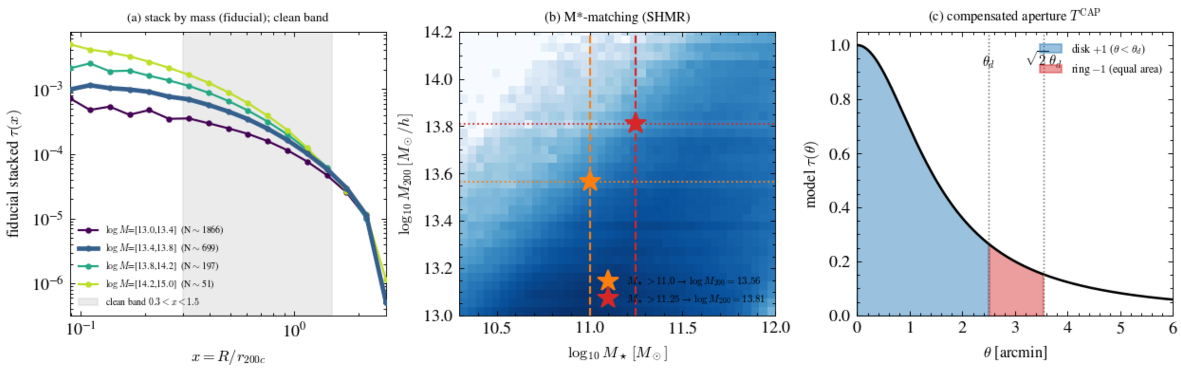

Figure 2: **(a)** Fiducial stacked $\tau(R/r_{200c})$ profile in four mass bins, with the "clean" fit band $0.3 < x < 1.5$ shaded. **(b)** The stellar-to-halo-mass relation used to place the two DESI BGS $M_\star$ cuts ($\log M_\star{>}11.0, 11.25$) onto host $\log M_{200c} = 13.56, 13.81$ respectively (the independent per-node mass estimate used for the Section 3.3 CAP-ratio confrontation gives the consistent $\log M_{200c} = 13.60, 13.82$; the ~ 0.01–0.04 dex difference reflects two separate SHMR-based reductions of the same $M_\star$ cuts, not a physical discrepancy). **(c)** The compensated-aperture-photometry (CAP) filter: a disk of radius θ_d $(+1)$ minus an equal-area ring out to $\sqrt{2}\,\theta_d$ (-1), applied identically to the painted τ and y maps and to the DESI×ACT data. Takeaway: this is the exact filter and exact host-mass matching used for every data confrontation in Section 3.3.

A retracted intermediate number, and the corrected reachable ceiling. An earlier pass through this analysis reported that the gas fraction "saturates at ∼41% of the cosmic value" and does not reach the eROSITA band under any single-parameter excursion. That number came from a different design – a 60-run one-at-a-time (OAT) sweep, not the 256-node Sobol suite used above – and from a raw, non-background-subtracted aperture estimator whose $f_{\rm gas}$ trivially plateaus toward the cosmic value $\Omega_b/\Omega_m = 0.159$ regardless of the underlying feedback, because the per-halo patches are projected through the full slab depth; the same working session that produced it flagged it explicitly as a line-of-sight projection artifact and retracted it. The corrected, background-subtracted OAT result is different in kind: at $z < 0.2$ ($M_{500c} \sim 1.3 \times 10^{13}\,{\rm M}_\odot/h$), the BIND fiducial gas fraction $f_{\rm gas,500} = 0.076$ agrees with the true TNG hydrodynamical value (the source session quantifies this only as "≈ TNG truth," not to a specific percentage[TODO: trace the exact BIND-vs-TNG-truth agreement at this mass bin/redshift if a tighter number is needed]) and with the mainstream X-ray relation of Eckert et al. (2019) ($f_{\rm gas,500} \approx 0.070$–$0.131$) to ∼9%, i.e. BIND/TNG is *not* anomalous relative to the mainstream X-ray literature. The tension is specifically with the contested eROSITA-Popesso et al. (2024) low value: the fiducial gas fraction is 2.9× the eROSITA-Popesso value at group scale, declining to ∼ 1.2× at cluster scale. The strongest single-parameter OAT excursion reaches $f_{\rm gas,500} = 0.047$, i.e. 1.8× the eROSITA-Popesso value – closing roughly 58% of the gap between the TNG fiducial and the eROSITA band under a single-knob excursion ($0.076 - 0.047 = 0.029$ of the $0.076 - 0.026 = 0.050$ total gap), but not reaching it. We report this corrected single-knob number, 0.047 (1.8× eROSITA, ∼58% of the gap closed), as the paper's reachable-ceiling statement in place of the retracted "41% cosmic" phrasing, and note explicitly that it comes from the earlier one-at-a-time design rather than the 256-node Sobol suite that produces the headline 0%-coverage result above. We were not able to confirm from the available sources whether the pre-computed $f_{\rm gas,500}$ column used directly for the 0%-coverage result of Figure 3 carries the identical line-of-sight/background-subtraction treatment applied in the corrected OAT pipeline; [TODO: verify parquet $f_{\rm gas,500}$ background-subtraction provenance against

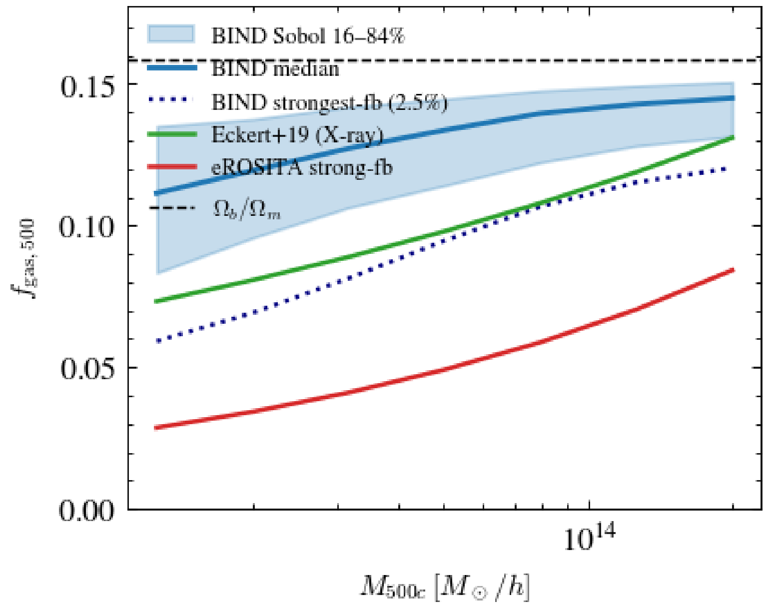

Figure 3: Gas fraction within $500c$ versus halo mass, for the 16–84% band, median, and strongest-feedback (2.5%) edge of the 256-node Sobol design, compared to the mainstream X-ray relation of Eckert et al. (2019) and the eROSITA strong-feedback edge of Popesso et al. (2024), over $M_{500c} = 10^{13}$–$3 \times 10^{14}\,\mathrm{M}_\odot/h$. The horizontal dashed line marks the cosmic baryon fraction Ω_b/Ω_m. Takeaway: *no* point in the 30-dimensional TNG-like feedback prior – including its most extreme feedback tail – reaches the eROSITA band; the entire Sobol volume sits above the strong-feedback curve.

the OAT reduction before final submission] – the two numbers ($f_{\mathrm{gas}} \approx 0.11$–$0.15$ at $M_{500c} \sim 13.5$ from the Sobol suite, versus 0.076 from the bg-subtracted single fiducial run at $z < 0.2$) are drawn from different mass bins and redshift windows and should not be read as contradicting one another.

3.2 τ/y CAP profiles versus the DESI×ACT GNFW fits

Figure 4 shows $\tau(R)$ and $y(R)$ for all 256 Sobol nodes in the BGS host mass bin, colored by the node's gas fraction $\tilde{f}_{\mathrm{gas,500}}$, with the fiducial TNG profile overlaid in black. Over the clean comparison band $x = R/r_{200c} \in [0.3, 1.5]$, BIND's τ-profile outer slope is consistent with the massive-BGS ($\log M_\star$>11) and ELG GNFW fits of Hadzhiyska et al. (2026) ($\beta \approx 4.8$–5.2 and $\beta \approx 4.5$ respectively), and shallower than the steep fit to the full BGS sample ($\beta_{\mathrm{BGS\_all}} = 7.3$); across the 256-node band, BIND spans $\beta \approx 4$–6 against the data's $\beta \approx 4.5$–7.3, an overlapping range. The GNFW inner-core parameter α is offset (BIND $\alpha \approx 1$–3 versus the data's $\alpha \approx 0.2$), but we flag this specific comparison as not physically meaningful: $\alpha \approx 0.2$ sits at a near-singular corner of the fixed-core ($x_c = 0.7$) GNFW form that is degenerate with the fixed inner slope, so the valid comparison is the projected-profile overlay and the outer slope β, not the (α, β) point itself. At fixed $\log M_{200c} \sim 13.6$, the pressure (y) response is higher at the ELG redshift ($z = 1.16$) than at the BGS redshift ($z = 0.18$) at fixed mass, as expected from the higher critical density $\rho_{\mathrm{cr}}(z)$; the implied electron-temperature proxy is ~ 1–2×10^7 K (~ 1–2 keV), declining with radius.

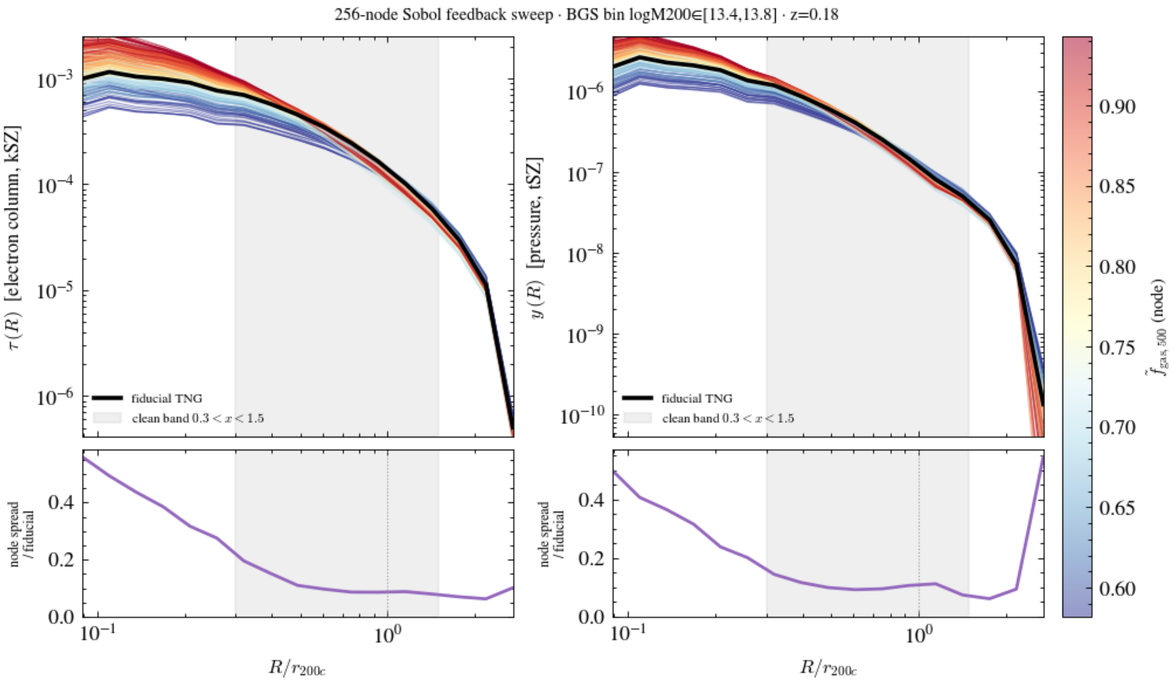

Figure 4: Stacked $\tau(R)$ (left) and $y(R)$ (right) profiles for all 256 Sobol feedback nodes in the BGS host mass bin ($\log M_{200c} \in [13.4, 13.8]$, $z = 0.18$), colored by each node's gas fraction $\tilde{f}_{\mathrm{gas},500}$; the fiducial TNG profile is overlaid in black. Bottom row: the ratio of the node-to-node spread to the fiducial profile as a function of radius, showing the feedback response is core-dominated. Takeaway: feedback reshapes both the τ and y profiles smoothly and coherently with $\tilde{f}_{\mathrm{gas}}$, most strongly at small radius.

3.3 The CAP-aperture reconciliation of the kSZ amplitude

The exact DESI×ACT observable is the CAP-ratio gas fraction $\tilde{f}_{\mathrm{gas}}(\theta_d)$ defined in Section 2.3. Figure 5 shows this quantity for the BIND fiducial run at the two DESI BGS $M_\star$ cuts, together with the 256-node Sobol band, against the real DESI×ACT data points. At the halo virial radius, the fiducial CAP gas fraction is $\tilde{f}_{\mathrm{gas}}(r_{200}) = 0.77$ for the $\log M_\star{>}11.0$ cut and 0.84 for the $\log M_\star{>}11.25$ cut, against measured data values of 0.44 and 0.60 respectively – ratios of 1.76× and 1.40×, at roughly 2σ significance. This is not, however, a wholesale exclusion of the design: 49 of the 256 Sobol nodes (at the $\log M_\star{>}11.0$ cut) and 65 of 256 (at the $\log M_\star{>}11.25$ cut) – roughly 20% of the design – are statistically consistent with the data within its (large) errors, and the single best-fitting node achieves $\chi^2 \approx 0.1$. The correct reading is therefore that the fiducial parametrization is too gas-rich by a factor of 1.4–1.8×, but the confrontation with the Sobol design as a whole is feedback- and data-limited, not a clean exclusion of every point in the design.

Retracted precursor claims. We explicitly do not use, and warn against reproducing, three earlier-stage numbers superseded by the result above: (i) a *cumulative*- (not compensated-) aperture version of $\tilde{f}_{\mathrm{gas}}$ that gave "0 of 256 nodes reach the data" and a ~ 2.5–$3.1\times$ tension, further inflated to $\sim 3\times$ before the angular-diameter-distance bug fix of Section 2.8; and (ii) a raw CAP-τ (not the $\tilde{f}_{\mathrm{gas}}$ ratio) amplitude comparison that suggested BIND is "2–6× above the data" at small aperture and "$\sim 1\times$" at $7.5'$ – this comparison was later shown to be normalization-dominated (folding in σ_v, the survey's velocity-reconstruction fidelity, T_{CMB}, and M_{200c} through a naive $\tau = T^{\mathrm{CAP}}/(T_{\mathrm{CMB}}\,\sigma_v/c)$

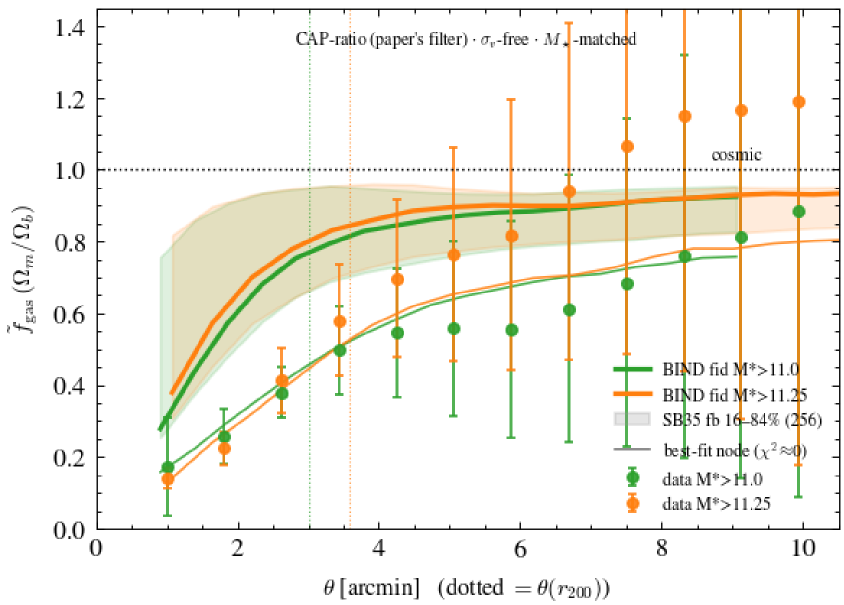

Figure 5: The σ_v-free, $M_\star$-matched CAP-ratio gas fraction $\tilde{f}_{\rm gas}(\theta)$ (Section 2.5) for the BIND fiducial run at the two DESI $M_\star$ cuts (green, orange), the 256-node Sobol 16–84% band, and the best-fitting single node (legend rounds its fit to $\chi^2 \approx 0$; the precise value is $\chi^2 \approx 0.1$, Section 3.3), against the real DESI×ACT data points with errors. The dotted vertical lines mark $\theta(r_{200})$ for each cut. Takeaway: the fiducial run is systematically above the data at small and intermediate aperture but roughly 20% of the Sobol design is statistically consistent with the data within its errors – this is the paper's corrected, final kSZ-amplitude statement, superseding all earlier (cumulative-aperture) CAP numbers.

conversion that is invalid) and was dropped from the paper entirely; the σ_v-free $\tilde{f}_{\rm gas}$ ratio of Figure 5 is the only CAP-amplitude number that appears in this paper.

The clean 3D capstone. Removing projection entirely, the per-halo integrated catalog (Section 2.2) gives a clean, unprojected 3D gas fraction $\tilde{f}_{\rm gas}(< r_{500}) = 0.82$ and $\tilde{f}_{\rm gas}(< r_{200}) = 0.88$ for the BIND fiducial run, against a DESI×ACT BGS value of ≈ 0.32 at the same aperture – BIND is $\sim 2.7\times$ too gas-rich within r_{200} in this projection-free comparison; the data only reach BIND's gas fraction at $\theta \approx 7$–$8'$ (~ 3–$4\ r_{200}$). This 3D number and the CAP-ratio number above are genuinely different observables (one projection-free, one an aperture-filtered projection) and are reported separately rather than combined.

The tension holds across mass and redshift. Extending the comparison to the DESI ELG sample at its own redshift ($z = 1.16$, using the patch-reuse technique of Section 4.1) and matching the data's $\sim 2.8\,r_{200}$ aperture reach, the fiducial BIND gas fraction is $\tilde{f}_{\rm gas}(2.8\,r_{200}) = 0.88$ (true-TNG value 0.77) against an ELG data value of ≈ 0.46 – a $\sim 2\times$ tension, indicating the gas excess persists from cluster/group scales at $z \sim 0.2$ through to the ELG regime at $z \sim 1.2$.

The mass anchor: it is gas, not mass. To rule out a halo-mass selection artifact as the source of the gas-fraction tension, we compare a CMB-lensing $\kappa(\theta)$ stack built from the same painted total-mass patches (ACT DR6 $\ell < 3000$ smoothing, background-subtracted, $M_\star$-matched) against the DESI BGS×ACT $\kappa(\theta)$ data. At the 1-halo scale ($\theta \sim 2.25'$), BIND's κ is $\sim 1.3\times$ the data – substantially smaller than the $\sim 2\times$ discrepancy seen in the (now-superseded, cumulative-aperture) $\tilde{f}_{\rm gas}$ metric – indicating BIND's halo masses are accurate to ∼30%, and that the $\tilde{f}_{\rm gas}$ tension is dominantly a gas-content effect, not a mass-selection artifact.

3.4 The first continuous-feedback kSZ+tSZ posterior

We trained the Gaussian-process emulator of Section 2.6 to map the 30 feedback parameters to the CAP-ratio amplitude at each aperture; its cross-validated $R^2 \approx 0.83$ at every aperture for the CAP amplitude, well above the $R^2 \approx 0.4$–0.7 achieved for shape-only quantities, confirming that the amplitude responds smoothly to feedback and validating the emulator for posterior inference. Sampling this emulator against a single BGS bin gives a data-limited posterior in which the most-constrained parameter, `UVBHepDeltaz`, is pinned to 0.86× its prior width, with no parameter strongly constrained. Jointly fitting both BGS $M_\star$ cuts ($\log M_\star$>11.0 and 11.25) modestly improves this: the best-constrained ratio tightens from 0.86 → 0.77× prior, and the number of parameters constrained below 0.85× prior grows from 0 to 5, pulling on the wind/supernova sector (`VariableWindVelFactor`, `WindEnergyIn1e51erg`, `WindFreeTravelDensFac`). Expanding to all five DESI BGS $M_\star$ cuts widens the constrained set further (0→6 parameters below 0.85× prior, minimum ratio 0.87 → 0.75), but adding the three ELG cuts (eight aperture points in total) barely moves the constraint further, consistent with the finding (Section 3.5) that the feedback response is intrinsically close to two-dimensional – once the two-dimensional manifold is reasonably sampled, additional data points along it are largely redundant. Restricting to the 49 nodes statistically consistent with the CAP-ratio data (Section 3.3), only the wind/supernova and IMF sector is pulled away from the prior, as a ∼6-knob combination (`VariableWindSpecMomentum`, `WindEnergyIn1e51erg`, `IMFslope`, `VariableWindVelFactor`, `WindFreeTravelDensFac`, `WindEnergyReductionFactor`, together with `RadioFeedbackFactor`, `BlackHoleFeedbackFactor`, `SNII_MinMass_Msun`, `RadioFeedbackReorientat` `MaxSfrTimescale`, and `QuasarThreshold`) rather than any single parameter moving in isolation.

Finally, we forecast a future joint kSZ+tSZ CAP analysis at Simons Observatory/CMB-S4×DESI sensitivity (Figure 6). Either probe alone leaves one feedback direction essentially prior-dominated (variance ratio ~ 0.9), but the joint kSZ+tSZ combination pins *both* directions: direction 1 to $\nu \approx 0.10$ and direction 2 to $\nu \approx 0.16$ (where ν is the ratio of posterior to prior variance), demonstrating that the two probes are complementary rather than redundant. Direction 1 loads positively on `VariableWindVelFactor` and `WindEnergyIn1e51erg` and negatively on `BlackHoleFeedbackFactor`; direction 2 loads on `SNIa_Rate`, `WindFreeTravelDensFac`, and `RadioFeedbackFactor`. The best-constrained forecast direction lies 0.81 inside the 2-D latent plane identified in Section 3.5 (compared to 0.26 for a random direction), i.e. the forecast pins essentially the same manifold that the current-data response already lives in, rather than a new, orthogonal direction.

3.5 Thermo decomposition: a 2-D inner/outer-gas latent, not a τ-vs-y split

A retracted 3-probe closure. An earlier version of this analysis attempted a direct thermodynamic decomposition of the form $y = \tilde{f}_{\rm gas} \cdot \kappa \cdot T_e$, reporting density (kSZ) 2.0×, mass (κ) 1.3×, and pressure (tSZ) 2.6× the data, implying an electron temperature within $\sim 1.0\times$ of the data ("too much gas, not too hot"). A same-day CAP-aperture revision updated the density leg to 1.7× and the implied T_e agreement to $\sim 1.2\times$ ("mostly too much gas, mildly too hot"), before the full closure was retracted the following day. This closure is retracted: the 2.6× tSZ number that drove the implied temperature agreement was itself the Msun-versus-Msun/h host-mass bug described in Section 2.8, and once that bug is fixed there is no longer a valid T_e/heating-versus-ejection decom-

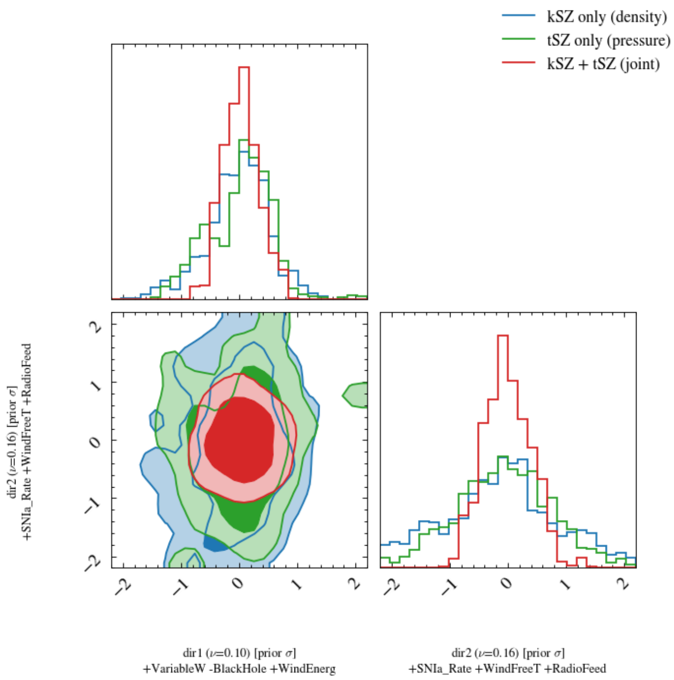

Figure 6: Forecast corner plot of the two feedback directions constrained by a future Simons Observatory/CMB-S4×DESI joint kSZ (density, blue) + tSZ (pressure, green) CAP analysis, compared to their joint combination (red). Takeaway: kSZ and tSZ alone each leave roughly one direction essentially unconstrained, but the joint combination pins both – the two probes are complementary, not redundant, and the pinned directions coincide with the 2-D feedback latent of Figure 8.

position to report; the cross-sample density×mass×temperature decomposition was removed from the analysis entirely.

What remains valid. The mass anchor of Section 3.3 (robust to the $Y \propto M^{5/3}$ bug because it uses total-mass patches, not Y) confirms BIND's masses are accurate to ∼30%, i.e. the tension is a density effect. The CAP-ratio gas fraction (density/kSZ leg) is 1.76×/1.40× the data at the two $M_\star$ cuts (Section 3.3). The corrected tSZ pressure leg (Figure 7), evaluated at the correct DESI-LRG lensing mass $\log M_{200c} = 13.18\,\mathrm{M}_\odot/h$ (Sailer et al., 2024), gives a fiducial y-CAP amplitude ≈ 1.51× the ACT×DESI-LRG y-CAP data (Liu et al., 2025), with the ±0.1-dex host-mass systematic *alone* spanning 1.35–2.05× – demonstrating that the tSZ amplitude comparison is mass-selection-dominated and not, by itself, a clean feedback probe. At this corrected mass, the kSZ and tSZ legs now *agree* at the strong-feedback edge of the response (maximum-a-posteriori $\hat{e}_1 \approx -7$, implied $\tilde{f}_{\rm gas} \approx 0.53$–$0.54$), rather than showing the inter-probe tension reported in the retracted closure above; a Spearman rank correlation of $\rho = 0.80$ between χ^2(kSZ) and χ^2(tSZ) across nodes confirms the same nodes fit both legs. (The direct visualization of the real data projected into the $(\hat{e}_1, \hat{e}_2)$ latent plane with kSZ-only/tSZ-only/joint posterior contours was cut from the figure set for space; the MAP and ρ values above are the quantitative record of that reconciliation.) The y-CAP *shape*, normalized at $2.25'$, is more radially extended in BIND than in the data (peaking at $4.75'$ versus the data's $3.5'$); we verified with a flat-pedestal test that this extension is a real feature of the profile, not a background-subtraction artifact, but – per the caveat discussed in Section 4.1 – we

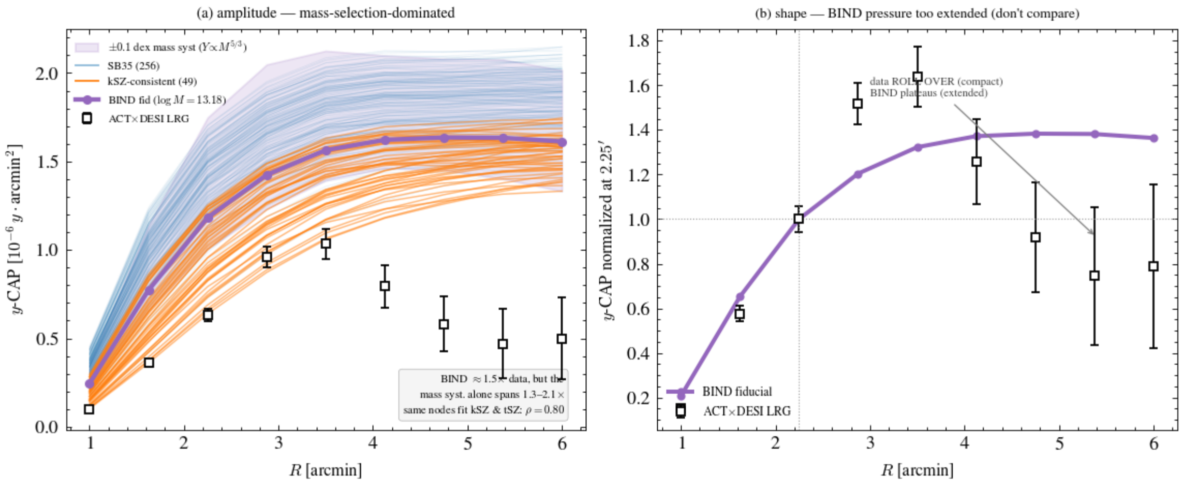

Figure 7: **(a)** y-CAP amplitude for all 256 Sobol nodes (light blue), the 49 kSZ-CAP-consistent nodes (orange), and the BIND fiducial run at the corrected DESI-LRG lensing mass $\log M_{200c} = 13.18\,\mathrm{M}_\odot/h$ (Sailer et al., 2024), against the ACT×DESI LRG data; the shaded band shows the ±0.1-dex host-mass systematic alone, which spans 1.35–2.05× and dominates the visual spread (the in-panel annotation rounds this to one decimal, 1.3–2.1×). **(b)** The y-CAP shape, normalized at 2.25′: BIND is more radially extended than the data (peak at 4.75′ versus 3.5′). Takeaway: the tSZ amplitude comparison is mass-selection-dominated – at the corrected mass, the kSZ and tSZ legs agree at the strong-feedback edge of the response, reversing the earlier (bug-driven) claim of an inter-probe tension.

do not read this shape comparison as a clean feedback diagnostic, since the tSZ stack is smoothed by miscentering, photometric-redshift line-of-sight scatter, and the ACT beam in ways the survey itself models and marginalizes but BIND does not attempt to reproduce.

The 2-D latent: inner gas versus outer gas. The paper's actual answer to "what drives the (τ, y) response" is not a τ-versus-y split but a *radial* one. Restricting to the clean band and the BGS mass bin, the 30-dimensional feedback response in $\log \tau$ and $\log y$ collapses onto just two latent components explaining 97% of the response variance ($\lambda = 0.51/0.46$, a sharp knee after the second component; Figure 8a). Rotating to physically interpretable axes, the first latent direction $\hat{e}_1$ correlates at $r = +0.95$ with the *inner*-gas fraction $f_{\rm gas}(< M_{500c})$ (the wind/supernova ejection axis), and the second, $\hat{e}_2$, correlates at $r = +0.95$ with the *outer*-gas fraction $f_{\rm gas}(M_{500c} \to M_{200c})$ (Figure 8c,d). These two radial zones are only weakly cross-correlated with each other ($r = 0.24$), i.e. they are genuinely independent degrees of freedom, and jointly reconstruct the 2-D latent plane with $R^2 = 0.90$ and 0.96 respectively. Total halo mass is essentially common-mode across nodes at fixed halo (coefficient of variation = 0.0008), so the response really is a statement about gas redistribution, not mass. An intermediate attempt to label the second axis with a gas-content-blind profile-concentration proxy was tested and found to be an overclaim – gas content and profile concentration turn out to be ∼87% degenerate in TNG, giving only a partial correlation of $r = 0.70$ – before the clean inner/outer gas-fraction labelling above was identified; we report only the final, physically validated labelling.

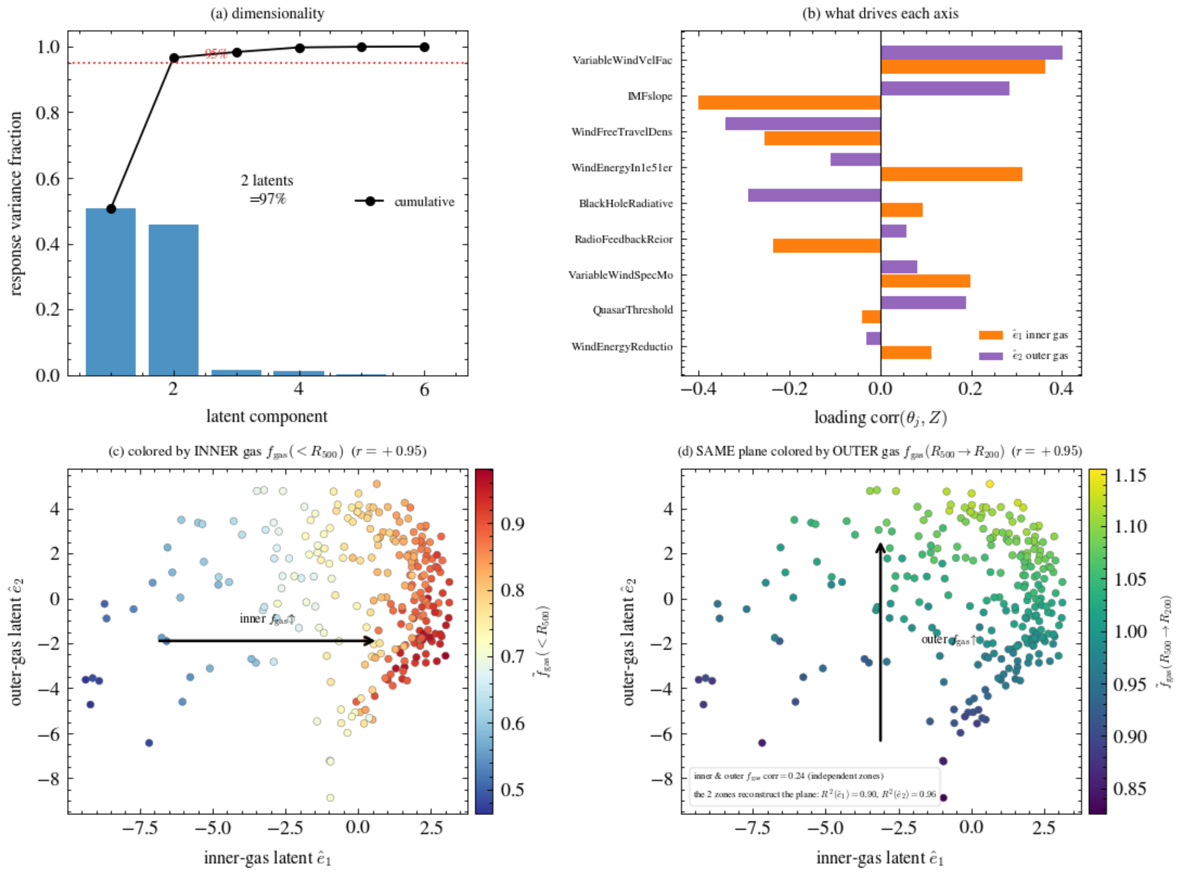

Figure 8: **(a)** Scree plot of the 30-parameter (τ, y) feedback response: two components explain 97% of the variance. **(b)** Parameter loadings on the two latent axes, dominated by `VariableWindVelFactor`, `IMFslope`, `WindFreeTravelDensFac`, and `WindEnergyIn1e51erg`. **(c,d)** The same 2-D latent plane colored by inner-gas $f_{\rm gas}(< M_{500c})$ (c) and outer-gas $f_{\rm gas}(M_{500c} \to M_{200c})$ (d); each latent axis correlates with its physical counterpart at $r = +0.95$. Takeaway: the response is a genuinely 2-D, radially-separated inner/outer gas manifold, not a single "more or less feedback" knob.

3.6 Field-level companion: the ray-traced multiprobe view

Sections 3.1–3.5 use painted patches stacked around individual halos; they never touch the ray-traced lightcone maps that the same suite also produces. As a deliberate companion analysis, we repeat the exercise using the assembled tomographic products of the ray-traced, 253-valid-node `emulator_dataset.npz` (angular power spectra $C_\ell^{\kappa\kappa}$, $C_\ell^{\kappa y}$, C_ℓ^{yy}, $C_\ell^{\tau\tau}$, the ready-made baryonic suppression $S(\ell, z_{\rm s})$ relative to the paired DMO lightcone (van Daalen et al., 2020), peak counts, PDFs, Minkowski functionals, and wavelet-scattering-transform coefficients) – no new reductions, only a different observable basis on the same underlying Sobol suite.

The field response is ∼1-D. In sharp contrast to the per-halo CGM's clean 2-D (inner/outer-gas) latent of Section 3.5, the field-level feedback response compresses onto essentially a single latent component: $\lambda_1 \approx 0.95$, robust across every probe combination tested (Figure 9). Our interpretation is that the line-of-sight projection and the lensing kernel wash out the radial second dimension that

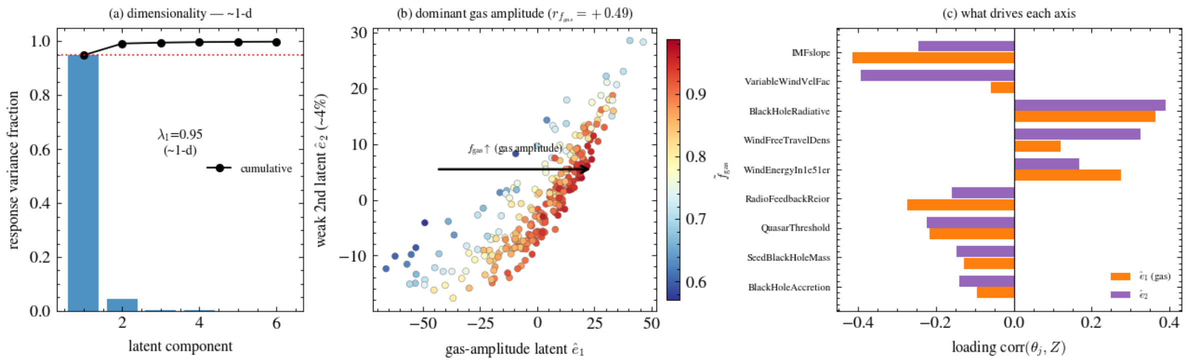

Figure 9: **(a)** Scree plot of the field-level $(S(\ell), C_\ell^{\kappa y})$ feedback response: component 1 already exceeds the 95%-variance line at $\lambda_1 \approx 0.95$, with component 2 essentially flat. **(b)** The dominant gas-amplitude latent versus the (weak) second latent, colored by $f_{\rm gas}$. **(c)** Parameter loadings on the two field-level axes. Takeaway: once projected through the lensing/tSZ kernel, the clean 2-D inner/outer-gas manifold of Figure 8 collapses to a single dominant gas-ejection direction.

is cleanly resolved in 3D around individual halos – the field-level observable is sensitive essentially only to a single gas-ejection amplitude.

Real-data confrontation: BIND shear$\times y$ versus DES Y3$\times$ACT. We compute the ray-traced $\xi_{\gamma y}(\theta)$ cross-correlation (via $n(z)$-weighting, the physical ACT DR6 beam of $2.4'$, a J_2 Hankel transform, and a robust angular window of $8'$–$40'$) and compare directly to the real DES Y3$\times$ACT measurement (Gatti et al., 2022; Pandey et al., 2022) (Figure 10). At well-measured scales, BIND is $\approx 2.5\times$ above the data for DES source bin 3 ($\bar{z} \approx 0.74$) and $\approx 2.3\times$ above the data for bin 4 ($\bar{z} \approx 0.94$), indicating the painted TNG-like gas is too strongly bound to halos in projection as well as in 3D. An earlier draft of this comparison used an erroneous, over-smoothing effective beam of $10'$, which spuriously shifted BIND's predicted peak to $11'$ against the data's peak at $4.5'$; replacing this with the correct, physical $2.4'$ ACT beam (caught by a coauthor during review) reveals that BIND is both too high in amplitude *and* too radially extended (too steep) relative to the data, consistent with the over-bound gas distribution inferred from the per-halo analysis above.

$\kappa\times y$, not κ auto, carries the feedback signal. Ranking the band-integrated feedback response D (the node-to-node spread relative to the fiducial, integrated over the ℓ range shown) across probe combinations gives $y\tau$ ($D = 0.18$) $>$ $\tau\tau$ (0.16) $>$ yy (0.135) $>$ κy (0.10) $\gg$ $\kappa\kappa$ (0.03): the weak-lensing auto-spectrum is comparatively feedback-blind ("mostly dark matter"), and the feedback signal lives predominantly in the pressure/density cross-correlations. A future LSST$\times$CMB-S4 $\kappa \times y$ forecast, with $5\times$ the Fisher information of the current DES$\times$ACT comparison, pins the single dominant field-level gas-ejection direction to $\nu \approx 0.05$, while the orthogonal direction remains comparatively wide at $\nu \approx 0.11$.

4 Discussion

Model-space versus parameter-space tension. The central result of this paper is Figure 3: across the full 30-dimensional IllustrisTNG-like feedback prior, sampled at 256 points including its strongest-feedback tail, not a single configuration reaches the eROSITA-Popesso strong-feedback gas-fraction band. Because this null result holds across the *entire* sampled parameter volume rather

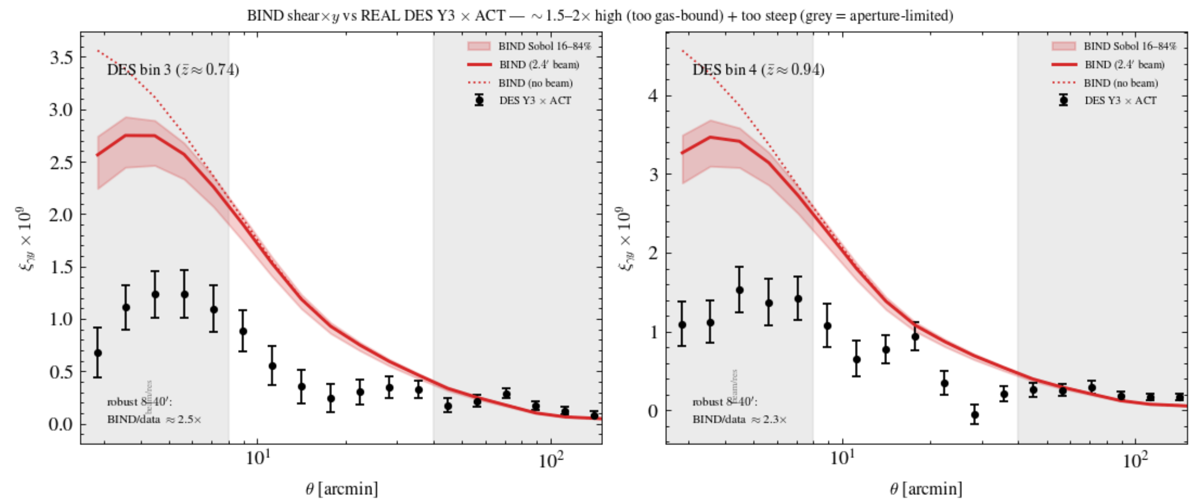

Figure 10: Ray-traced BIND shear×y cross-correlation $\xi_{\gamma y}(\theta)$ (16–84% Sobol band, 2.4′-beam-convolved solid, no-beam dotted) compared to the real DES Y3×ACT measurement (Gatti et al., 2022; Pandey et al., 2022) for DES source bins 3 and 4. The shaded regions outside the robust 8′–40′ window are affected by the finite (5°) field low-ℓ cut. Takeaway: at robust scales, BIND is $\approx 2.5\times/2.3\times$ above the real data in bins 3/4 – confirming, at the field level and against real survey data, the same gas-excess tension identified per-halo in Section 3.3. (The panel's own super-title rounds this to a stale "$\sim$ 1.5–2×" estimate from an earlier reduction pass; the precise, current values are the in-panel-annotated and body-text 2.5×/2.3× quoted here.)

than a single fiducial point, it cannot be resolved by recalibrating TNG's feedback parameters within the model as currently formulated – it is a statement about the reachable range of the TNG-like feedback *model space*, not about an unlucky choice of parameters within it. At the same time, the CAP-ratio comparison of Section 3.3 shows the confrontation with the DESI×ACT kSZ data specifically is less absolute: roughly 20% of the design is statistically consistent with the CAP-ratio data given its current errors, so the two datasets (eROSITA and DESI×ACT kSZ) currently constrain the TNG-like model space with different strength, and the eROSITA tension is the more robust of the two as things stand.

What would reach the band? The 2-D inner/outer-gas latent of Section 3.5 identifies the wind/supernova and IMF sector as the combination that moves the response furthest toward the data-consistent region, but even the strongest single-parameter excursion tested closes only ~58% of the gap to the eROSITA band (Section 3.1). This suggests that reaching the eROSITA-Popesso regime would require either a qualitatively different feedback mechanism than TNG's wind/AGN parametrization provides – for instance, gas ejection efficient enough to push baryons well beyond M_{200c} rather than merely redistributing them between M_{500c} and M_{200c} – or some other change to the underlying subgrid model rather than a parameter adjustment within it. We do not have a direct test of this hypothesis in the current suite and flag it as the natural next step.

The velocity caveat, and the path to a true $\Delta T_{\rm kSZ}$. Every kSZ-labelled comparison in this paper uses the velocity-free electron column τ (Section 2.7), not a reconstructed temperature decrement. This is an important limitation: real stacked-kSZ measurements are sensitive to the product of gas density and line-of-sight peculiar velocity, and a genuine $\Delta T_{\rm kSZ}$ comparison could in principle differ from the τ-only comparison presented here if BIND's implicit gas kinematics (inherited from the painted density field alone, without a dedicated velocity model) differ systematically from the true TNG gas velocity field. Building a literal $\Delta T_{\rm kSZ}$ observable would require a dark-matter-only velocity surrogate for the gas peculiar-velocity field, which is not yet built in this suite (a planned Phase-3 extension, gated on this kill criterion, was not attempted here).

Selection effects. Three selection-related caveats bear directly on the results above and are discussed further in Section 4.1: the painted-halo mass floor systematically under-samples the DESI ELG host population; the tSZ y-CAP profile shape (though not its amplitude) is compared only qualitatively because of miscentering and beam effects the survey itself models but BIND does not reproduce; and the CAP posterior covariances used in Section 3.4 are approximate (diagonal across nested $M_\star$ bins for the CAP posterior; a simplified AR(1) aperture covariance for the multiprobe forecast), both flagged as open methodological improvements rather than results.

4.1 Caveats

We collect here, as instructed by the analysis plan, the caveats that qualify every result above.

- **Velocity-free τ.** BIND's "kSZ" throughout this paper is the electron-column optical depth, not a velocity-weighted temperature decrement; a literal $\Delta T_{\rm kSZ}$ is unbuilt and would require a dark-matter-only velocity surrogate (Section 4).

- **TNG model space only.** The 256-node Sobol design samples TNG-*like* feedback parametrizations; it cannot speak to non-TNG subgrid physics, and the paper's central claim (Section 4) is explicitly about this model space, not a universal statement about all possible galaxy-formation models.

- **Mass floor of $10^{13}\,\mathrm{M}_\odot/h$ for painted halos.** The TNG300 lightcone halo catalog floor is $M_{200c} \geq 10^{13}\,\mathrm{M}_\odot/h$; the DESI ELG-host regime sits below it. A low-mass (10^{12}–$10^{13}\,\mathrm{M}_\odot/h$) patch-reuse technique recovers only a partial sample: 28.7% of the target population within the production $4 \times R_{200}$ circular paste, and 51.6% within the full 6.25 Mpc/h patch footprint. The uncaptured $\sim$48% are field halos too far from any $\geq 10^{13}\,\mathrm{M}_\odot/h$ cluster to be reused (captured halos have a median host distance of 1.87–1.9 Mpc/h, i.e. they are systematically near-cluster rather than field halos), though an independent environment-bias check against the full TNG300 hydrodynamical catalog found the captured sample's gas fraction is nonetheless representative of this particular statistic (near-host versus field $\tilde{f}_{\rm gas} = 0.408$ versus 0.409, a $< 0.3\%$ difference).

- **Satellite/miscentering effects in CAP stacks.** The tSZ y-CAP profile *shape* (not amplitude) is not treated as a clean feedback diagnostic because the real photometric-LRG stack is additionally smoothed by miscentering, photometric-redshift line-of-sight scatter, and the ACT beam, effects the survey models and marginalizes but that BIND does not attempt to reproduce; for the same reason, the CMB-lensing κ comparison of Section 3.3 is restricted to the 1-halo scale ($\theta \lesssim 3'$), since BIND's isolated per-halo patches carry no 2-halo term.

- **y-map normalization.** The painted Compton-y maps are already physical and require no additional $1/a^2$ redshift weighting, consistent with this project's standing normalization convention, previously validated against Planck.

- f_{gas} **provenance gap.** We could not confirm from the available sources whether the pre-computed $f_{\text{gas},500}$ column used directly for the 0%-coverage headline result of Section 3.1 carries the same line-of-sight background-subtraction treatment later applied in the corrected one-at-a-time reduction. [TODO: resolve before final submission]

- **Compare like with like.** BIND's f_{gas} is total gas mass; eROSITA measures hot, X-ray-emitting gas only; kSZ is sensitive to all free electrons. The X-ray-based band should be read as a lower envelope on the true gas content, not a literal kSZ target.

- **Central stellar-mass proxy.** The painted central stellar-mass proxy used to match halos to DESI $M_\star$ cuts has not yet been independently validated against TNG Subfind stellar masses at the fiducial parameter point (an open methodological TODO).

5 Conclusions

Painting the BIND emulator's hydrodynamical fields onto a shared population of TNG300 lightcone halos under a 256-node Sobol sampling of the 30-dimensional IllustrisTNG feedback prior lets us ask, for the first time in this context, whether *any* TNG-like feedback configuration reproduces the strong gas depletion reported by recent DESI×ACT kSZ and eROSITA analyses. The answer is no: zero of the 256 sampled feedback configurations – including the design's most extreme feedback tail – reach the eROSITA strong-feedback gas-fraction band at group/cluster scale. Against the DESI×ACT kSZ CAP-ratio observable specifically, the tension is less absolute (the fiducial run is 1.4–1.8× too gas-rich, but roughly 20% of the design is statistically consistent with the data within its current errors), and a CMB-lensing mass anchor confirms the discrepancy is predominantly a gas-content, not a mass-selection, effect. The 30-dimensional per-halo (τ, y) feedback response reduces to a clean, physically interpretable 2-D latent – independent inner- and outer-gas fraction axes – while the same feedback response, viewed through a ray-traced lensing/tSZ kernel, compresses to a single dominant gas-ejection direction, a concrete illustration of how projection can hide the dimensionality of the underlying physics. We provide, for the first time in this suite, per-halo posteriors on the feedback parameters directly from (τ, y) CAP stacks and a σ_v-free gas-fraction estimator that sidesteps the survey's velocity calibration entirely. The overall picture supports reading the missing-baryon tension as a statement about the reachable range of TNG-like feedback *models*, not as evidence that the public IllustrisTNG parameter choice happens to be miscalibrated within an otherwise-adequate model. Closing this gap – either by identifying a qualitatively different feedback mechanism or by building the dark-matter-only velocity surrogate needed for a literal ΔT_{kSZ} comparison – is the natural next step for this suite.

Acknowledgments

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